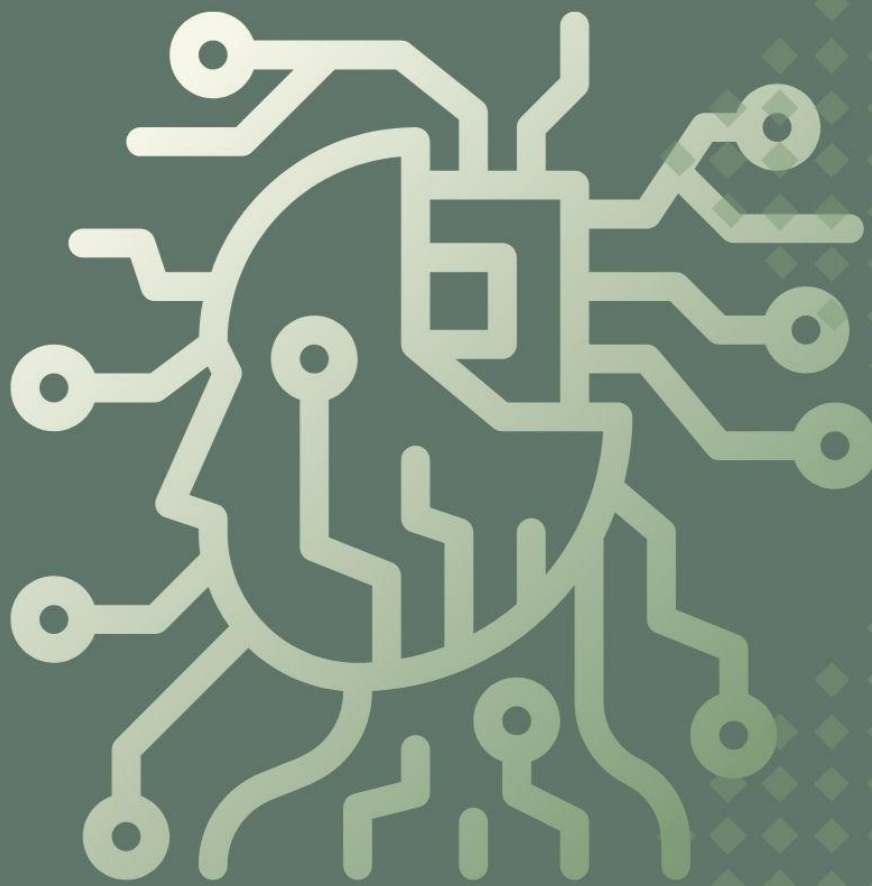




WIRED TO WRITE: TECHNICAL WRITING INSTRUCTIONAL MATERIAL

for Information Technology, Entertainment and Multimedia
Computing, Automotive Technology, and Electronics
Technology Students

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MODULE 2

Lesson 2: Effective Technical Writing: Research, Documentation, and Ethical Practices

Specific Learning Objective:

Discuss credible academic and industry sources into technical documents using standardized citation formats.

Illustrate project or experimental findings in a clear, concise, and ethically responsible manner.

INTO

Activity 1: Source Scavenger Hunt

Instruction: You will work in pairs to complete a "Source Scavenger Hunt." Each pair will find one academic source, such as a journal article, and one industry source, like a report or website, that is relevant to your field. For each source, list the title, author, and publication details. Then, identify why each source is credible. Consider factors such as the author's qualifications, whether the work is peer-reviewed, and its timeliness. Finally, explain how each source could be used in a technical document, such as supporting an argument or providing background information.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

THROUGH

Activity 2: Reading

Instructions: Read the provided texts on the importance of utilizing credible sources, standard citation methods, and illustrating findings clearly. Highlight key points and discuss how you can apply these techniques to your own writing.

Technical writing is a crucial skill for students in fields such as Information Technology, Automotive Technology, Electronics Technology, and Entertainment and Multimedia Computing. This lesson will help you understand the importance of using credible sources and proper documentation techniques when writing technical documents, and it will introduce you to the core principles of presenting findings ethically and clearly.

1. Credible Sources in Technical Writing

When writing a technical document, it is critical to use credible sources. Credibility is key because the information you present must be reliable and trustworthy. Credible sources usually come from reputable authors, institutions, or organizations. Academic sources, such as peer-reviewed journal articles, are considered credible because experts in the field have reviewed them. Industry sources, such as reports from reputable companies or technical manuals, are also crucial in technical writing, as they provide practical, real-world information.

To determine if a source is credible, ask yourself these questions:

- ✓ Is the author an expert in the field?
- ✓ Is the source published by a reputable organization or journal?
- ✓ Is the information up-to-date and relevant?
- ✓ Is the source free from biases or conflicts of interest?

The following are some of the credible online sources in your field:

A. Google Scholar

A freely accessible web search engine that indexes scholarly articles across various disciplines, including computer science, engineering, and multimedia studies.

Access: <https://scholar.google.com/>

☰ My profile ★ My library

Google Scholar

Recommended articles

☆

HIGHER EDUCATION REVENUE & EMPLOYABILITY
P Aliphat, N Damjanovic, P Tapie
files.osf.io - 23 days ago

PDF

☆

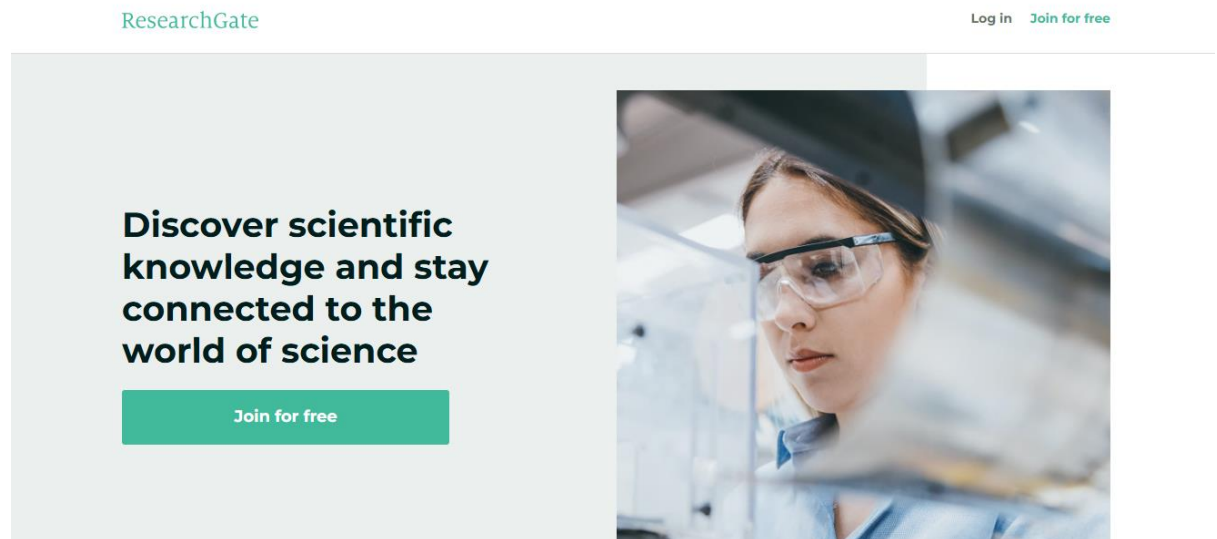
TECHNICAL-PRODUCTIVE EDUCATION AND THE
EMPLOYABILITY OF GRADUATES: A SYSTEMATIC
REVIEW
R Asto, G Maritsa
Aula Virtual - 30 days ago

These recommendations are based on your [Scholar profile](#) and alerts.

B. ResearchGate

A professional network for researchers to share papers, ask and answer questions, and find collaborators in fields like electronics, automotive engineering, and multimedia computing.

Access: <https://www.researchgate.net/>



C. IEEE Xplore Digital Library

A leading digital library for research in electrical engineering, electronics, and computer science, providing access to journals, conferences, and standards.

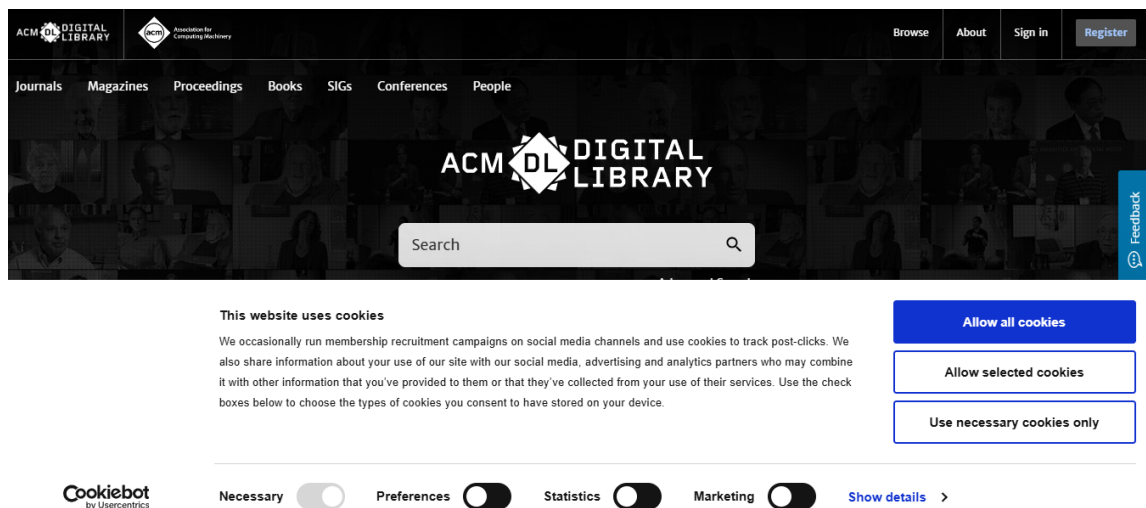
Access: <https://ieeexplore.ieee.org/Xplore/home.jsp>



D. ACM Digital Library

A comprehensive digital library for computing and information technology, offering access to journals, conference proceedings, and technical magazines.

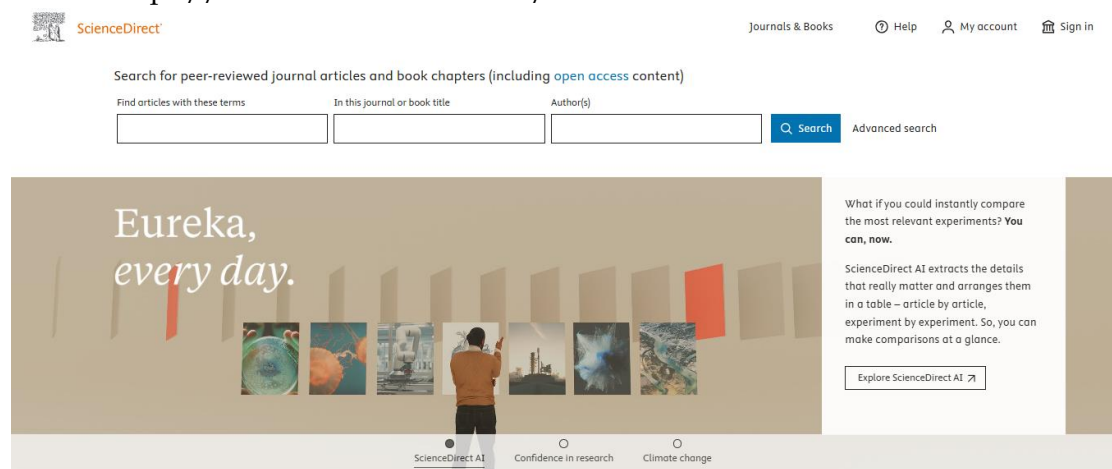
Access: <https://dl.acm.org/>



E. ScienceDirect

A leading full-text scientific database offering journal articles and book chapters from more than 2,500 peer-reviewed journals in science and engineering.

Access: <https://www.sciencedirect.com/>

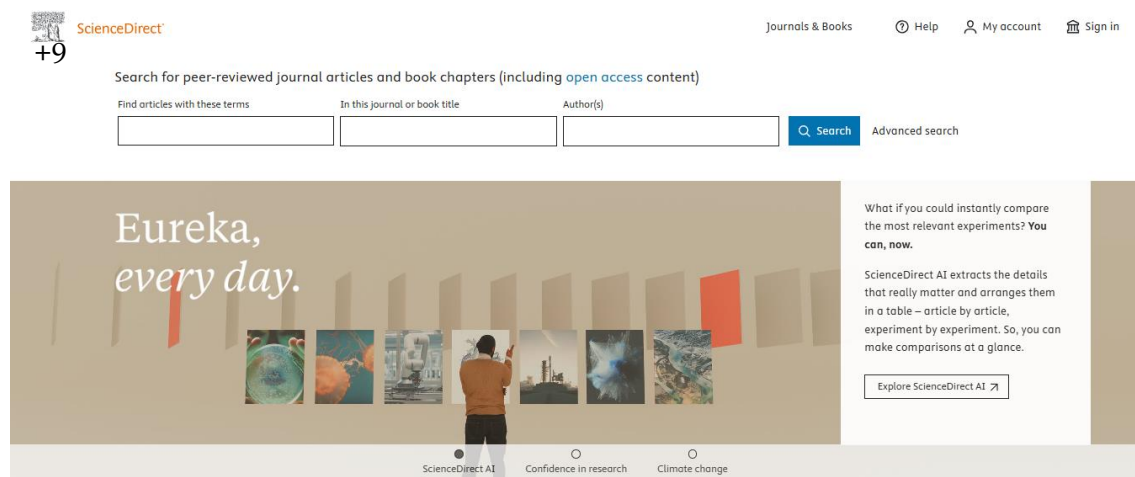


Explore scientific, technical, and medical research on ScienceDirect

F. JSTOR

A digital library for scholars, researchers, and students, providing access to thousands of academic journals, books, and primary sources across various disciplines.

Access: https://www.jstor.org/?utm_source=chatgpt.com

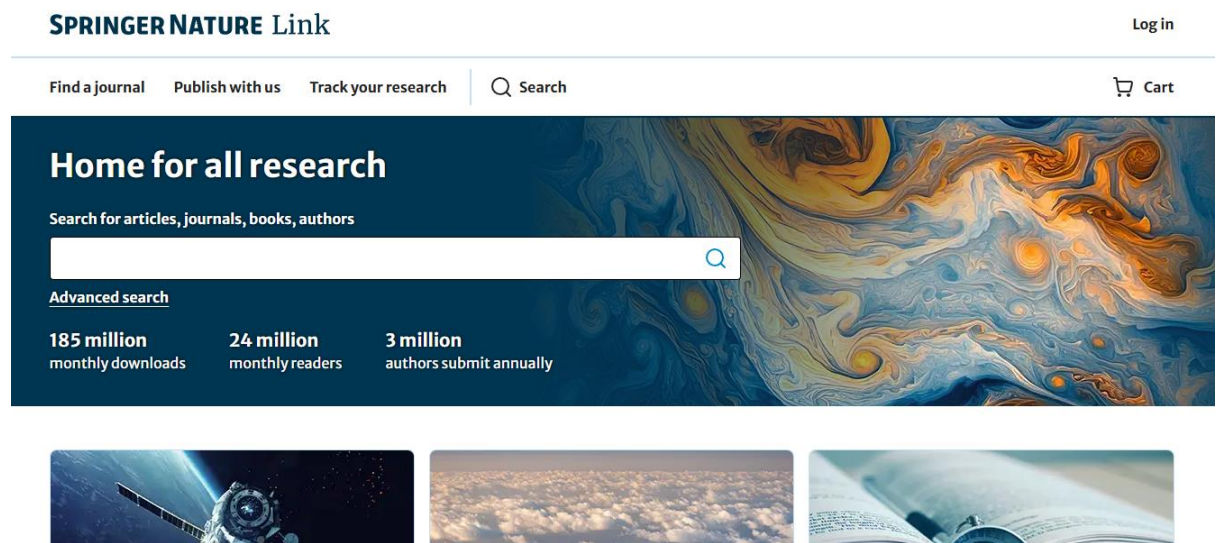


Explore scientific, technical, and medical research on ScienceDirect

G. SpringerLink

Provides access to millions of scientific documents from journals, books, series, protocols, reference works, and proceedings in fields like electronics and engineering.

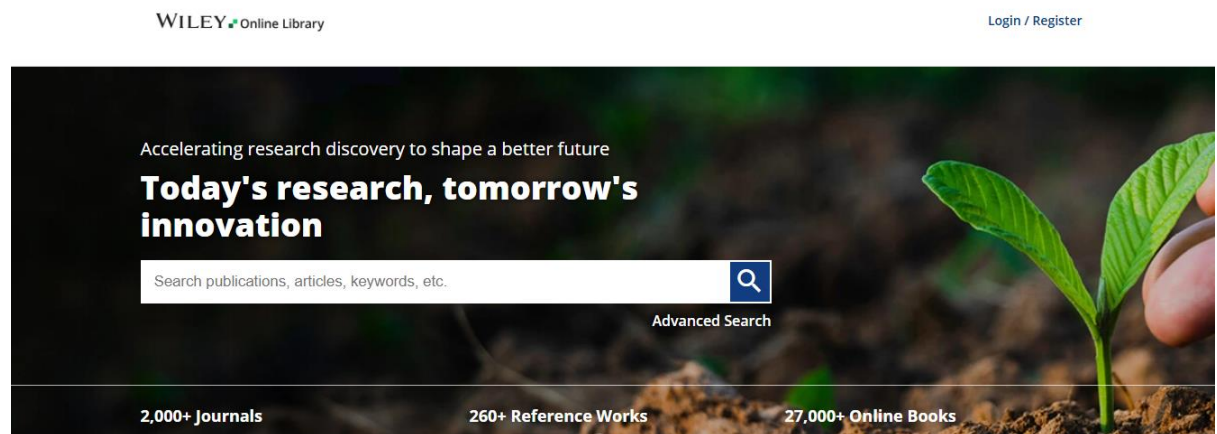
Access: <https://link.springer.com/>



H. Wiley Online Library

A vast multidisciplinary collection of online resources covering life, health, and physical sciences, including engineering and technology.

Access: <https://onlinelibrary.wiley.com/>

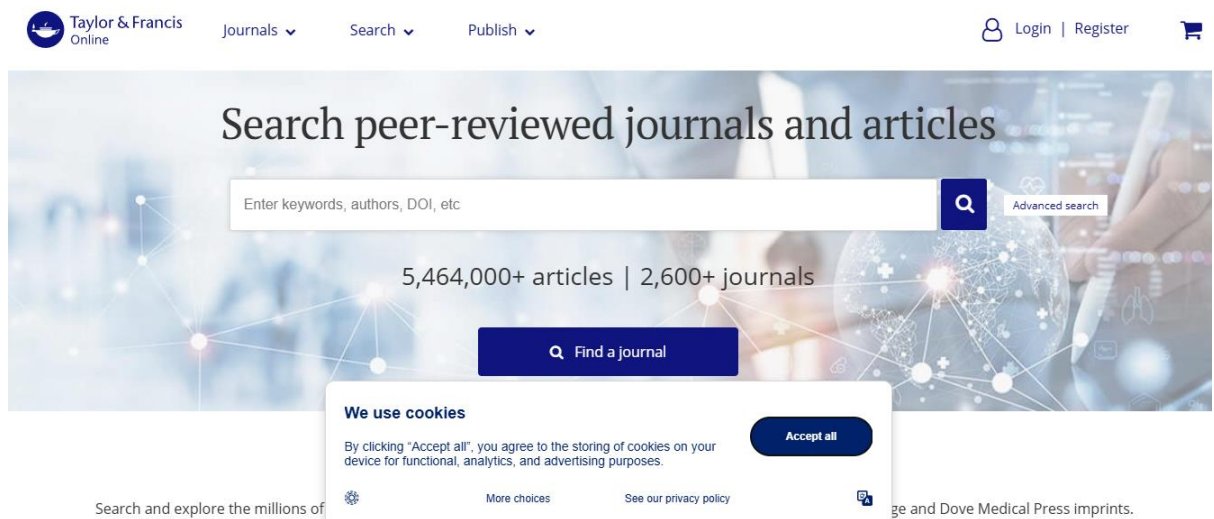


Resources

I. Taylor & Francis Online

Hosts a wide range of academic journals in science, technology, and engineering, offering access to peer-reviewed research articles.

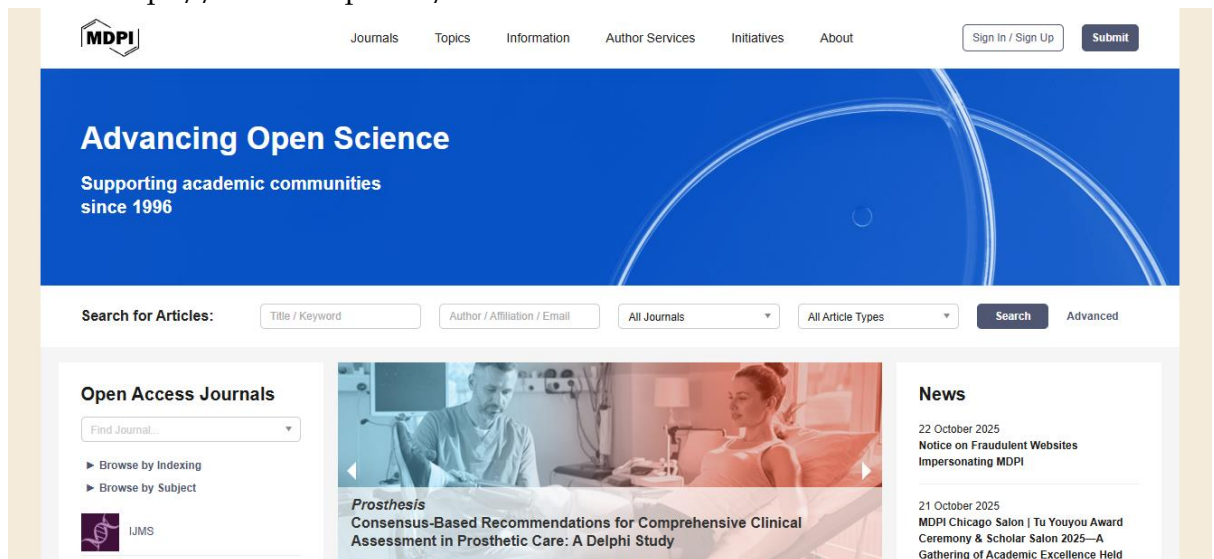
Access: <https://www.tandfonline.com/>



J. MDPI

An open-access publisher offering peer-reviewed journals in various disciplines, including engineering, technology, and multimedia computing.

Access: <https://www.mdpi.com/>



2. Standard Citation Formats

Proper citation of sources is essential in technical writing. Using standardized citation formats, such as APA, MLA, or Chicago, helps give credit to the original authors and provides readers with the ability to verify the information. APA 7th Edition is the most commonly used citation style for technical writing, providing guidelines for citing books, articles, websites, and other sources. The following are 5 of the most common citation formats and their proper citations. Observe the appropriate formatting, including capitalization, italization, indentation, and punctuation marks.

A. APA (American Psychological Association) – 7th Edition

APA style is commonly used in the social and behavioral sciences, including psychology, education, and social work. It emphasizes clarity, precision, and the use of credible sources.

- Journal Article:

Template:

Author(s). (Year). Title of the article. *Title of the Journal*, Volume (Issue), page range.
<https://doi.org/DOI>

Example:

Smith, R. T. (2020). Introduction to automotive engineering. *Journal of Engineering Studies*, 45(2), 123-135. <https://doi.org/10.1016/j.jes.2020.01.005>

- Book:

Template: Author(s). (Year). *Title of the book*. Publisher.

Example:

Smith, R. T. (2020). *Introduction to automotive engineering*. Wiley.

- Website:

Template:

Author(s). (Year, Month Day). Title of the webpage. *Website Name*. URL

Example:

Kramer, S. (2022, March 15). Advances in electronics: The rise of 5G technology. *Electronics Today*. <https://www.electronicstoday.com/advances-in-5g>

B. MLA (Modern Language Association) – 9th Edition

MLA style is widely used in the humanities, particularly for literature, history, and cultural studies. It emphasizes authorship and clarity in citing sources.

- Journal Article:

Template:

Author's Last Name, First Name. "Title of Article." *Title of Journal*, vol. Volume, no. Issue, Year, pp. page range. <https://doi.org/DOI>

Example:

Smith, Robert T. "Introduction to Automotive Engineering." *Journal of Engineering Studies*, vol. 45, no. 2, 2020, pp. 123-135.
<https://doi.org/10.1016/j.jes.2020.01.005>

- Book:

Template:

Author's Last Name, First Name. *Title of Book*. Publisher, Year.

Example:

Smith, Robert T. *Introduction to Automotive Engineering*. Wiley, 2020.

- Website:

Template:

Author's Last Name, First Name. "Title of Webpage." *Website Name*, Day Month Year, URL

Example:

Kramer, S. "Advances in Electronics: The Rise of 5G Technology." *Electronics Today*, 15 Mar. 2022, <https://www.electronicstoday.com/advances-in-5g>.

C. Chicago Manual of Style – 17th Edition

The Chicago Manual of Style offers two systems: notes and bibliography (for humanities) and author-date (for sciences). It is flexible and widely used in various academic and professional fields.

- Journal Article:

Template:

Author's Last Name, First Name. "Title of Article." *Title of Journal* Volume, no. Issue (Year): page range. <https://doi.org/DOI>

Example:

Smith, Robert T. "Introduction to Automotive Engineering." *Journal of Engineering Studies* 45, no. 2 (2020): 123-135. <https://doi.org/10.1016/j.jes.2020.01.005>.

- Book:

Template:

Author's Last Name, First Name. *Title of Book*. Publisher, Year.

Example:

Smith, Robert T. *Introduction to Automotive Engineering*. Wiley, 2020.

- Website:

Template:

Author's Last Name, First Name. "Title of Webpage." *Website Name*, Month Day, Year. URL

Example:

Kramer, S. "Advances in Electronics: The Rise of 5G Technology." *Electronics Today*, March 15, 2022. <https://www.electronicstoday.com/advances-in-5g>.

D. Harvard

Harvard referencing is commonly used in the UK and other countries. It focuses on author-date citation, where the publication year is included in the text, making it suitable for a variety of academic disciplines.

- Journal Article:

Template:

Author's Last Name, Initial(s). (Year) 'Title of Article', *Title of Journal*, Volume(Issue), pp. page range. Available at: URL [Accessed: Day Month Year].

Example:

Smith, R. T. (2020) 'Introduction to automotive engineering', *Journal of Engineering Studies*, 45(2), pp. 123-135. Available at: <https://doi.org/10.1016/j.jes.2020.01.005> [Accessed: 20 October 2023].

- Book:

Template:

Author's Last Name, Initial(s). (Year) *Title of Book*. Publisher.

Example:

Smith, R. T. (2020) *Introduction to automotive engineering*. Wiley.

- **Website:**

Template:

Author's Last Name, Initial(s). (Year) 'Title of Webpage', *Website Name*. Available at: URL [Accessed: Day Month Year].

Example:

Kramer, S. (2022) 'Advances in electronics: The rise of 5G technology', *Electronics Today*. Available at: <https://www.electronicstoday.com/advances-in-5g> [Accessed: 20 October 2023].

E. IEEE (Institute of Electrical and Electronics Engineers)

IEEE style is commonly used in engineering and computer science disciplines. It uses numerical references in the text, where sources are listed in the order they appear in the document.

- **Journal Article:**

Template:

[#] Author(s) Initial(s). Surname(s), "Title of the article," *Abbreviated Title of Journal*, vol. Volume, no. Issue, pp. page range, Month Year. [Format]. Available: URL. [Accessed: Day Month Year].

Example:

[1] R. T. Smith, "Introduction to automotive engineering," *Journal of Engineering Studies*, vol. 45, no. 2, pp. 123-135, Feb. 2020. [Online]. Available: <https://doi.org/10.1016/j.jes.2020.01.005>. [Accessed: Oct. 20, 2023].

- **Book:**

Template:

[#] Author(s) Initial(s). Surname(s), *Title of Book*, Publisher, Year.

Example:

[2] R. T. Smith, *Introduction to automotive engineering*, Wiley, 2020.

- **Website:**

Template:

[#] Author(s) Initial(s). Surname(s), "Title of the webpage," *Website Name*, Month Day, Year. [Online]. Available: URL. [Accessed: Day Month Year].

Example:

[3] S. Kramer, "Advances in electronics: The rise of 5G technology," *Electronics Today*, Mar. 15, 2022. [Online]. Available: <https://www.electronicstoday.com/advances-in-5g>. [Accessed: Oct. 20, 2023].

3. Illustrating Project Findings Clearly and Ethically

When presenting experimental findings or technical data, it is essential to be clear, concise, and ethical. Technical writing often involves sharing complex data and results, so it's crucial to organize your findings in a way that is easy for your audience to understand. Avoid jargon and unnecessary details, and focus on the core message. Ethical writing also means presenting data honestly, without manipulation or omission of essential facts, to ensure that the information you present is truthful and transparent.

Best practices for illustrating findings include:

- ✓ Be clear and concise: Focus on key points and avoid overcomplicating your explanations.
- ✓ Use visuals when appropriate: Diagrams, charts, and graphs can help make complex information easier to digest.
- ✓ Provide context: Explain the significance of your findings and how they relate to the field or industry.
- ✓ Be ethical: Always represent data accurately and avoid exaggerating results to fit a desired outcome.

4. Applying Ethical and Clear Presentation in Technical Documents

Finally, it is essential to ensure that your technical documents follow both ethical and clarity standards. The purpose of technical writing is to communicate complex ideas, so being transparent with your methods, findings, and conclusions will help your readers trust the content.

For example, when writing an experimental report, you should:

- ✓ Provide a detailed methodology: How did you experiment? What tools and materials did you use?
- ✓ Present your results accurately: Use tables or graphs to present data, and include proper citations for any research or data sources.
- ✓ Conclude responsibly: Summarize your findings and discuss their relevance without overstepping or making unsupported claims.

In this lesson, we've discussed the importance of credible sources, the role of citation in technical writing, and how to illustrate findings clearly and ethically. These skills are essential for producing high-quality technical documents that make a meaningful contribution to your field. Remember, good technical writing is not just about the information you present, but how you present it—clearly, ethically, and with respect for your audience.

Activity 3: Creating an Annotated Bibliography

Instructions:

1. Select an *existing* sample project or research paper: Choose a sample research paper, project, or topic relevant to your field of study. You can select an article, book, or website.
2. Create Citations in APA 7th Edition and IEEE Format: For each source, create a citation in APA 7th Edition format and IEEE citation format as discussed in the previous activity.
3. Write the Annotated Bibliography: After each citation, write a summary (approximately 150 words) of the source that includes:
 - Summary of the Content: Briefly describe what the source is about.
 - Relevance to Your Work/Field: Explain how this source contributes to your research or project.

- Evaluation: Critique the source in terms of credibility, authority, and usefulness. Was it peer-reviewed? Is the author credible? Is the information up-to-date and relevant to your work?
4. Include In-Text Citations: Ensure that you have incorporated in-text citations for each source used, correctly formatted in both APA 7th Edition and IEEE.
 5. Peer Feedback: After completing your annotated bibliography, exchange it with a classmate for feedback. Review each other's work and suggest improvements in citation format or annotation clarity.
 6. Be guided by the rubric provided in the next page.

[illegible]

Annotated Bibliography Rubric:

Criteria	Excellent (4)	Proficient (3)	Basic (2)	Needs Improvement (1)	Points
Citation Formatting (APA 7th Edition & IEEE)	Both citations are perfectly formatted in APA 7th Edition and IEEE with no errors.	Citations are mostly correct, with minor formatting errors in one or both styles.	Citations have several errors or missing elements in one or both styles.	Citations are largely incorrect or missing for both styles.	6
Summary of Source & Relevance to Research	Provides a thorough, accurate summary of the source with a clear, well-explained relevance to the research/project.	Provides a clear summary with good relevance to the research/project, but could be slightly more detailed.	Summary is basic or unclear; relevance to the project is not fully explained.	Summary is too brief, unclear, or irrelevant to the project.	10
Evaluation of Source & In-Text Citation	Insightful critique of the source, evaluating its authority, credibility, and relevance. In-text citations are correctly integrated in both APA and IEEE.	Good critique with some insight; in-text citations are mostly correct, with minor mistakes.	Critique is minimal or lacks depth; in-text citations are mostly incorrect or missing.	No critique or evaluation; in-text citations are missing or incorrect.	7
Clarity, Organization, and Writing Quality	Well-organized and easy to follow; writing is clear, professional, and free from errors.	Organization is good, but some parts could be clearer; few grammatical or spelling errors.	Organization is unclear in parts; multiple grammatical or spelling errors.	Disorganized or hard to follow; many errors that impact readability.	7
Total Score: _____					

BEYOND

Activity 4: Experimental Findings Audio Log

Instructions: You are to create a 3–5-minute audio log, summarizing a technical project or experiment they’ve worked on. They must include the findings, the methods used, and properly cite any sources referenced using APA 7th edition. The audio should be clear, well-structured, and free from unnecessary jargon; you should also focus on maintaining ethical responsibility in your reporting. Submit this online.

Be guided by the rubric on the next page.

Experimental Findings Audio Log Rubric:

Criteria	Excellent (5)	Proficient (4)	Basic (3)	Needs Improvement (2)	Points
Clarity and Structure	Audio is exceptionally clear and well-structured; the narrative is logical and easy to follow.	Audio is clear and structured well, but some parts may require more clarity or better organization.	Audio lacks clarity in some parts, making it harder to follow the main points.	Audio is unclear, poorly structured, and difficult to follow; major issues with flow.	10
Content and Summary of Findings	Findings are presented in a detailed, thorough, and well-organized manner, covering all essential details.	Findings are presented clearly, with good detail, but may miss minor aspects or lack depth.	Findings are presented, but some key points are missing or underexplained.	Findings are vague, lacking key information, or poorly explained.	12
Methodology and Experiment Description	Methodology is thoroughly explained, including tools, techniques, and reasoning behind choices.	Methodology is well explained, though some details may be lacking or unclear.	Methodology is only partially explained; lacks details or clarity on certain steps.	Methodology is poorly explained or missing key components; unclear or inadequate.	8
Ethical Responsibility and Accuracy	All findings are reported honestly, and there's a clear focus on ethical responsibility in the reporting.	Findings are reported accurately, but minor concerns regarding ethical considerations may be present.	Some findings lack clarity or are inaccurately presented; limited focus on ethical responsibility.	Findings are presented dishonestly, or ethical responsibility is not considered.	7
Citation of Sources (APA 7th Edition)	Sources are properly cited in APA 7th Edition, both verbally and in text, without error.	Most sources are cited correctly in APA 7th Edition, but minor mistakes in citation or phrasing.	Some sources are cited in APA 7th Edition, but several errors in citation formatting.	No sources are cited, or citations are completely incorrect and fail to follow APA guidelines.	5
Presentation and Voice Quality	Voice is clear, professional, and engaging throughout the audio log; good pacing and	Voice is clear, but may lack engagement or have slight issues with pacing or tone.	Voice is somewhat difficult to understand; pacing or tone is inconsistent, or	Voice is unclear, difficult to understand, or the pacing/tone distracts from the message.	5

Criteria	Excellent (5)	Proficient (4)	Basic (3)	Needs Improvement (2)	Points
	appropriate tone.		the audio is monotone.		
Conciseness and Avoidance of Jargon	The audio is concise, with no unnecessary information or jargon. Language is simple, clear, and accessible.	The audio is mostly concise but may include some unnecessary details or jargon that distract from the message.	Audio contains some unnecessary details or technical jargon that may confuse the listener.	Audio is overly wordy or filled with jargon, making it hard for the listener to understand the key points.	3
Engagement and Communication	The student engages the listener, making the content interesting and easy to follow; clear enthusiasm.	The content is clear and informative, though it may lack engagement or enthusiasm.	The content is informative, but lacks engagement or the student sounds disinterested or disengaged.	The content is dull, and the student struggles to maintain the listener's interest.	3
Adherence to Time Limit (3-5 minutes)	Audio is between 3-5 minutes and covers all necessary points concisely within the time limit.	Audio is within the time limit, though some points may be rushed or overexplained.	Audio is under or over time, missing some essential points or spending too much time on others.	Audio is either too short or too long, affecting the clarity and structure of the presentation.	5
Total Score: _____					